Apart from apex predators, all animals must balance the need to sleep, reproduce, and feed with the risks of being eaten. The lethal role of predators is obvious, but their nonlethal (nonconsumptive) role can have an even bigger impact on an ecosystem. Potential prey are forced to change their way of life in order to avoid being killed.

In 1990, Werner studied the effects of green darner dragonfly larvae on toad tadpoles. He noticed that when the predatory larvae were in the tank, the tadpoles were less active, swam to other parts of the tank, and metamorphosed into adults when they were smaller. The predator had changed the toads' morphology and their behavior just by being there.

In 1991, Werner investigated what happened when more than one prey species was involved. In the absence of a predator, bullfrog and green frog tadpoles grew at virtually identical rates. However, when predatory dragonfly larvae were introduced to the tank, both prey species became less active and chose different places in which to swim. The bullfrog tadpoles grew more quickly than they had in a predator-free tank, but the green frog tadpoles decreased their feeding activity and grew more slowly. Werner concluded that for prey species there was a trade-off between the need to grow as fast as possible and the risk of predation. Growing more quickly requires more feeding activity, and this in turn increases the chances of being eaten by a predator. As the larvae's presence altered the behavior of the prey species differently, the bullfrog's new behavior gave it a competitive advantage over the green frog by making it bigger.